UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 8A – Work Area Design

EIN 3241: Human Factors and Ergonomics
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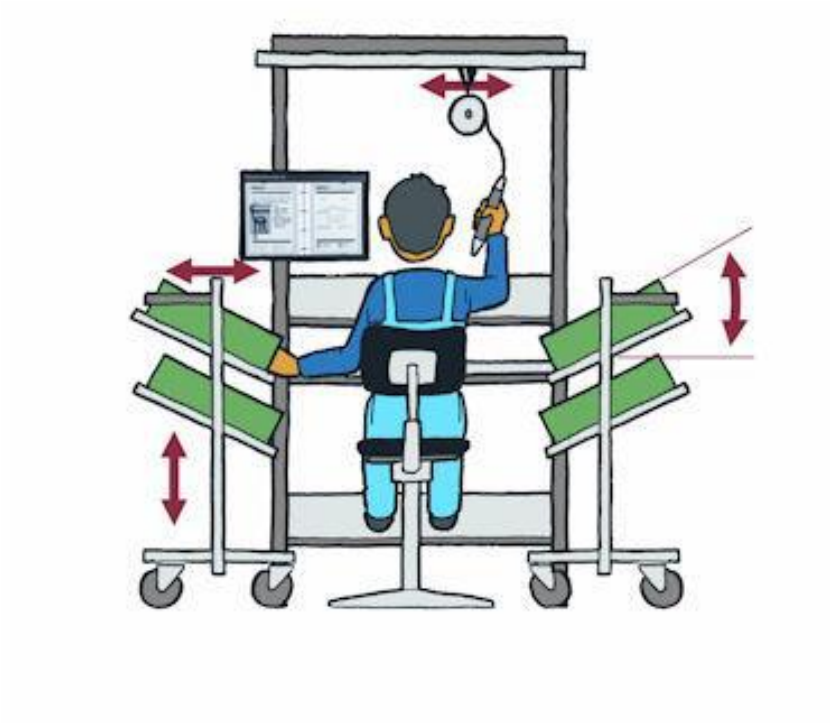
Admin Issues

- Last Module!
- Peer Evaluations

Introduction

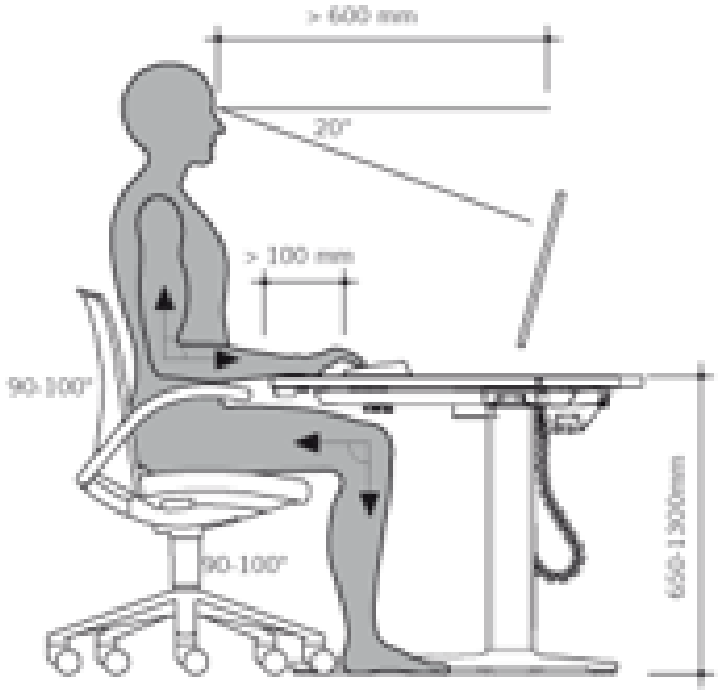
- **Workstation** is where one operator works with one or more pieces of equipment.



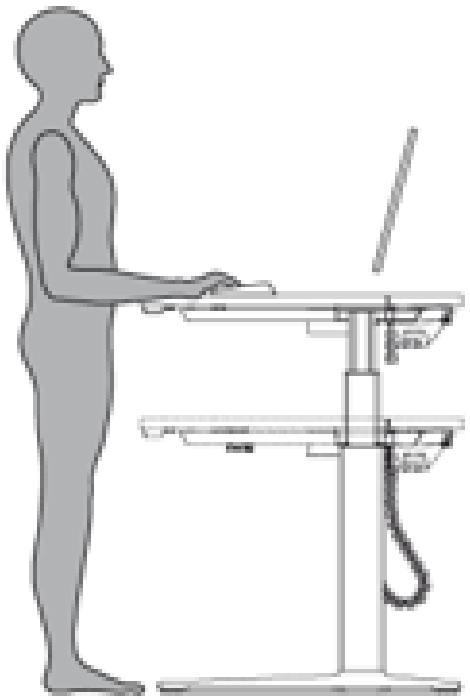
Postures at Work

- Workstations are generally either:

SEATED WORKSTATIONS



STANDING WORKSTATIONS



Seated Work Areas

- A seated posture:
  - Improves feeling of balance
  - Allows for reduced static load to maintain body postures
  - Improves blood circulation
  - Less fatigue development
- However, long periods of sitting on a poorly designed chair may lead to:
  - Slackening of abdominal muscles
  - Back and neck aches
  - Curvature of the spine

Standing Work Areas

- A standing posture:
  - Greater physiological load, maintaining balance and posture
  - May lead to blood and body fluid accumulation in legs (if not moving).
- So, which design is better standing or seated work areas?

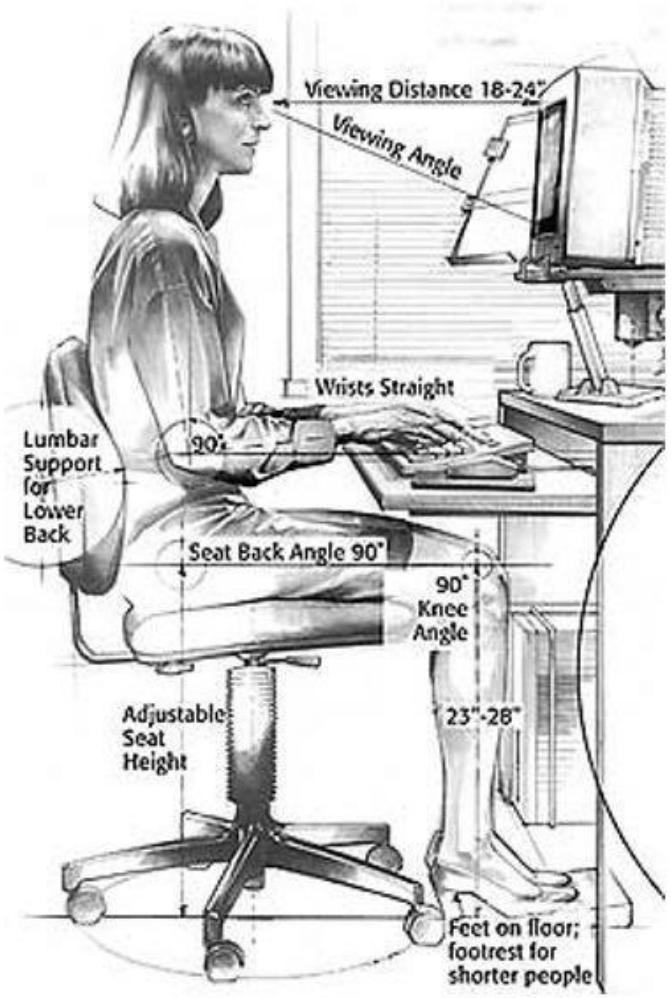
SEATED vs STANDING

- Exercise: In each of the below cases, which type of workstation is more suitable:

1. Precise foot control actions are required.
2. Fine assembly or writing tasks are predominant.
3. There are frequent demands for downward force application (wrapping, packing).
4. Heavy objects are being handled [more than 4.5 kg (10 lb.) in weight].
5. No leg and knee clearance is available.
6. No large forces are required.
7. A high degree of body stability or equilibrium is required.
8. Mobility is a must.
9. Long work periods are expected.
10. All items needed within the task cycle can easily be supplied within the seated workspace.
11. High, low, or extended reaches are frequently required.

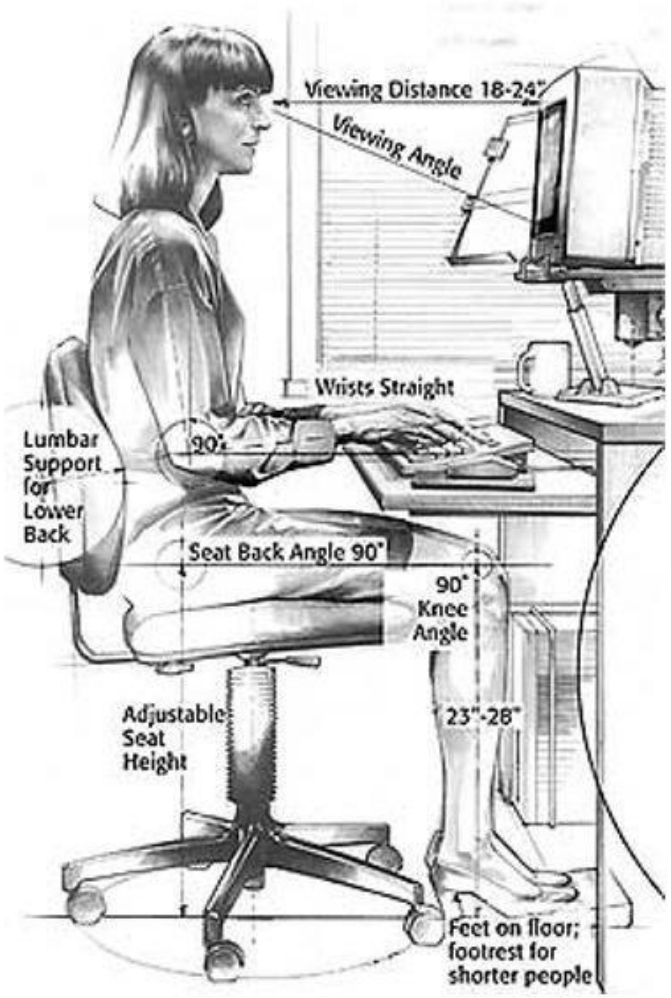
SEATED Work Areas

- Design considerations for seated work areas:
  - Correct eye position relative to viewing tasks
  - Adjustability of seat height.
  - Adjustable of back rest (if small).
  - Backrest width.
  - Lumbar support provided.
  - Seat surface depth and width.
  - Seat surface sloped backward with a slight dip in the middle for extra body stability.
  - Front edge of seat should be curving downwards.
  - Cushioned seat.



SEATED Work Areas (Cont'd)

- Design considerations for seated work areas:
  - Clearance for the thighs and the knees
  - Foot rests provided.
  - Arm support for reducing intervertebral disk pressure. If trunk mobility is desired, armrests are not recommended; pads on work surface may be considered instead.
  - Place larger and heavier objects closer to the operator.
  - Work surface height depends on elbow height and the nature of the task.
  - Multiple legs on seat for stability.



STANDING Work Areas

- Design considerations for standing work areas:
  - Proper monitor position with respect to NLS
  - Reach distances
  - Knee and foot clearance
  - A smooth and level standing surface must be provided.
  - There should be room for occasional moving around.
  - A nonslip and padded standing surface (floor mat) is recommended.
  - If two hand reaches are required, design for shorter reach distances.
  - Work surface height depends on the task being performed.

Guiding principles of arrangement

- *Importance principle*. Components that are most important in the functioning of the system are located in the optimal locations. Rate each component's criticality in meeting system's objectives and assign locations accordingly.
- *Sequence of use principle*. Review sequential relationships between components under different tasks. Those always used one after the other would be located next to each other.
- *Functional principle*. Components that are functionally related to each other are grouped together. Then groups and components within groups can be arranged following other principles.
- *Frequency of use principle*. Components that are used most frequently are located in the most accessible/optimal areas.

Work Area Arrangements: Link Analysis

- Link analysis: an objective method of developing data for component arrangement.
  - A *sequential link* of 5 for A B means that component B has been used five times right after component A.
  - A *functional link* reports the number of times a component is used per unit of time or a task cycle.

Example 1

- Determine the **sequential links** between components.

Process	# of times per day	Sequence
1	30	A-C-B-D-E
2	12	A-B-D-E
3	14	A-C-D-B-E

From \ To	A	C	B	D	E
A					
C					
B					
D					
E					

Example 1

Step-by-step breakdown:

- Process 1 (30 times/day): A → C → B → D → E
  - A → C: +30
  - C → B: +30
  - B → D: +30
  - D → E: +30

- Process 2 (12 times/day): A → B → D → E
  - A → B: +12
  - B → D: +12
  - D → E: +12

- Process 3 (14 times/day): A → C → D → B → E
  - A → C: +14
  - C → D: +14
  - D → B: +14
  - B → E: +14

1. Understand Workflow Relationships
2. Guide Facility Layout / Workstation Arrangement
3. Improve Task Design & Efficiency
4. Basis for System Optimization

From \ To	A	C	B	D	E
A		44	12		
C			30	14	
B				42	14
D			14		42
E					

Example 2

- Determine the **functional links** between components.

Process	# of times per day	Sequence
1	12	C-B-D-E
2	30	A-B-D-E
3	14	A-C-B

Example 2

- Step 1: List All Component Pairs per Process (unordered)
  - **Process 1 (12 times/day): {C, B, D, E}** → All combinations:
  - C–B
  - C–D
  - C–E
  - B–D
  - B–E
  - D–E
  - → Add 12 to each

  - **Process 2 (30 times/day): {A, B, D, E}** → All combinations:
  - A–B
  - A–D
  - A–E
  - B–D
  - B–E
  - D–E
  - → Add 30 to each

  - **Process 3 (14 times/day): {A, C, B}**→ All combinations:
  - A–C
  - A–B
  - B–C
  - → Add 14 to each
- Step 2: Sum the Frequencies for Each Pair

Example 2 (cont'd)

Component Pair	Total Functional Link
A–B	30 + 14 = **44**
A–D	30
A–E	30
A–C	14
B–C	12 + 14 = **26**
B–D	12 + 30 = **42**
B–E	12 + 30 = **42**
C–D	12
C–E	12
D–E	12 + 30 = **42**

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LEADING THE CHARGE, CHARGING AHEAD